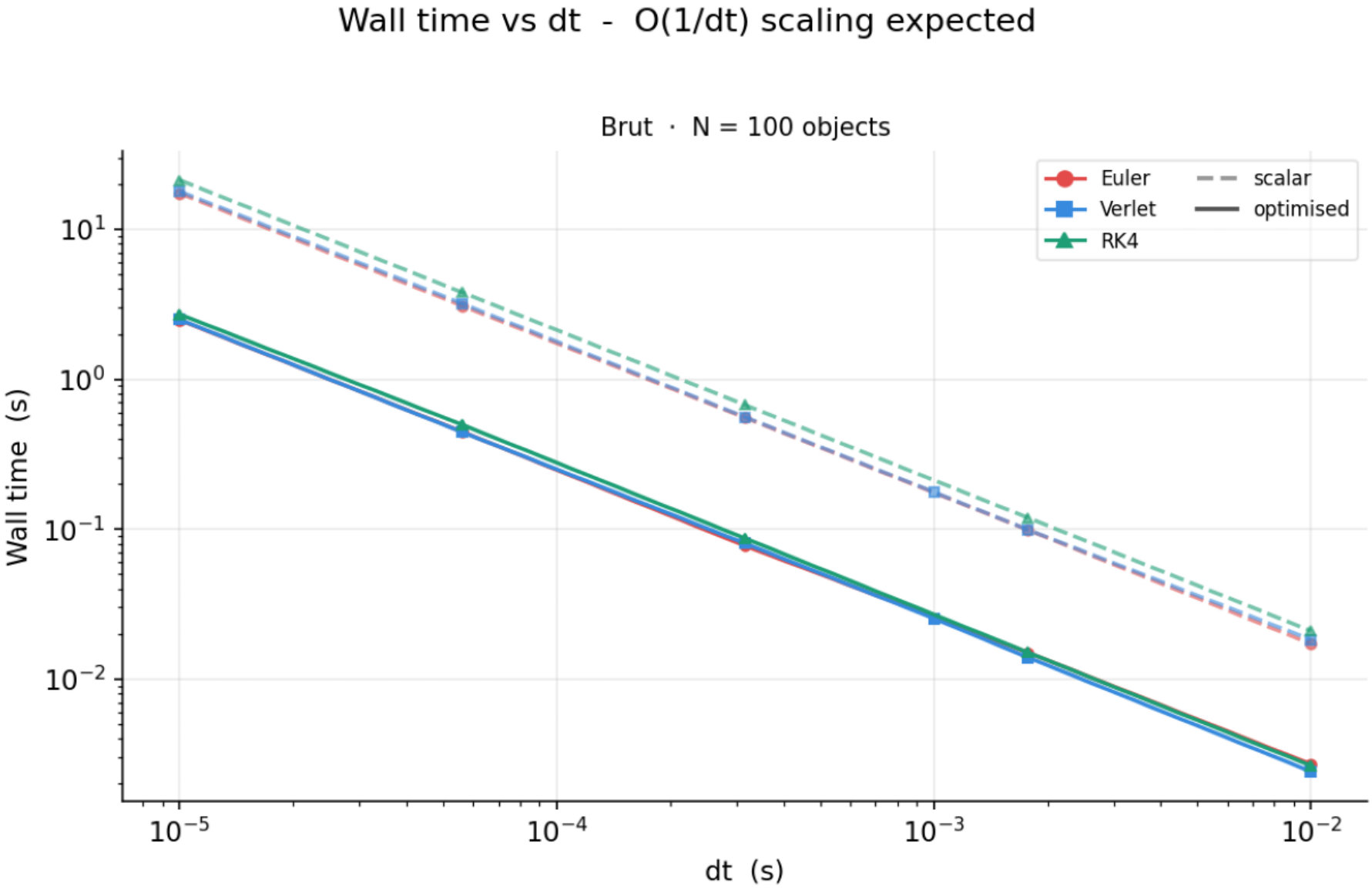
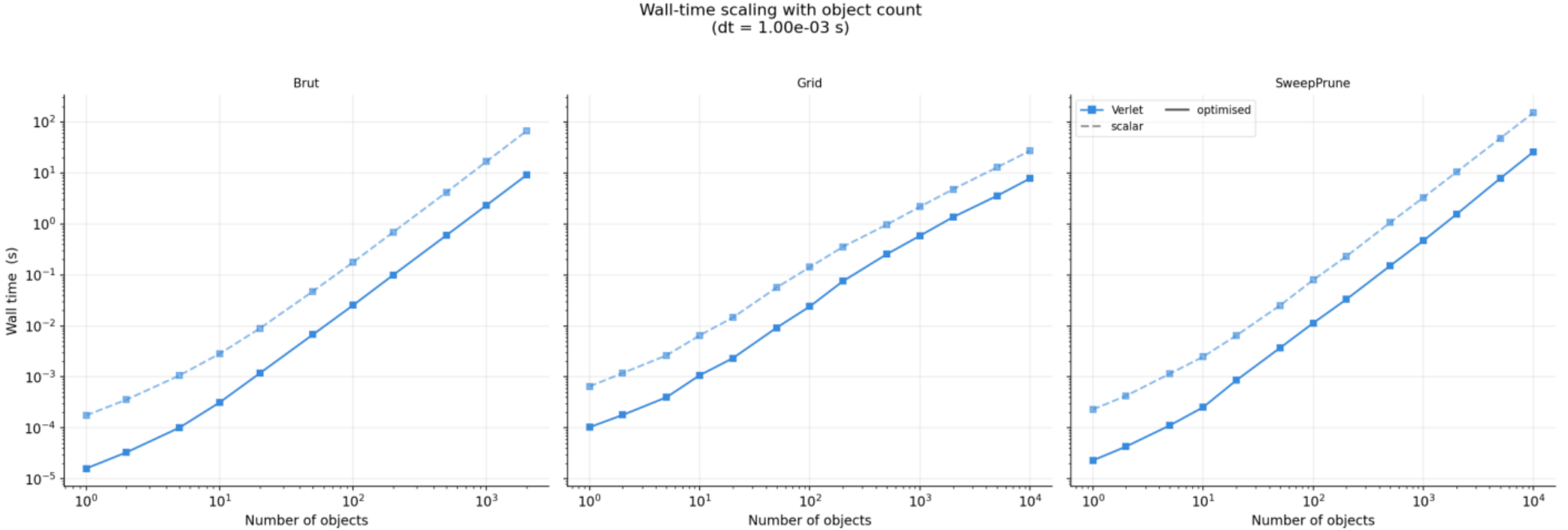
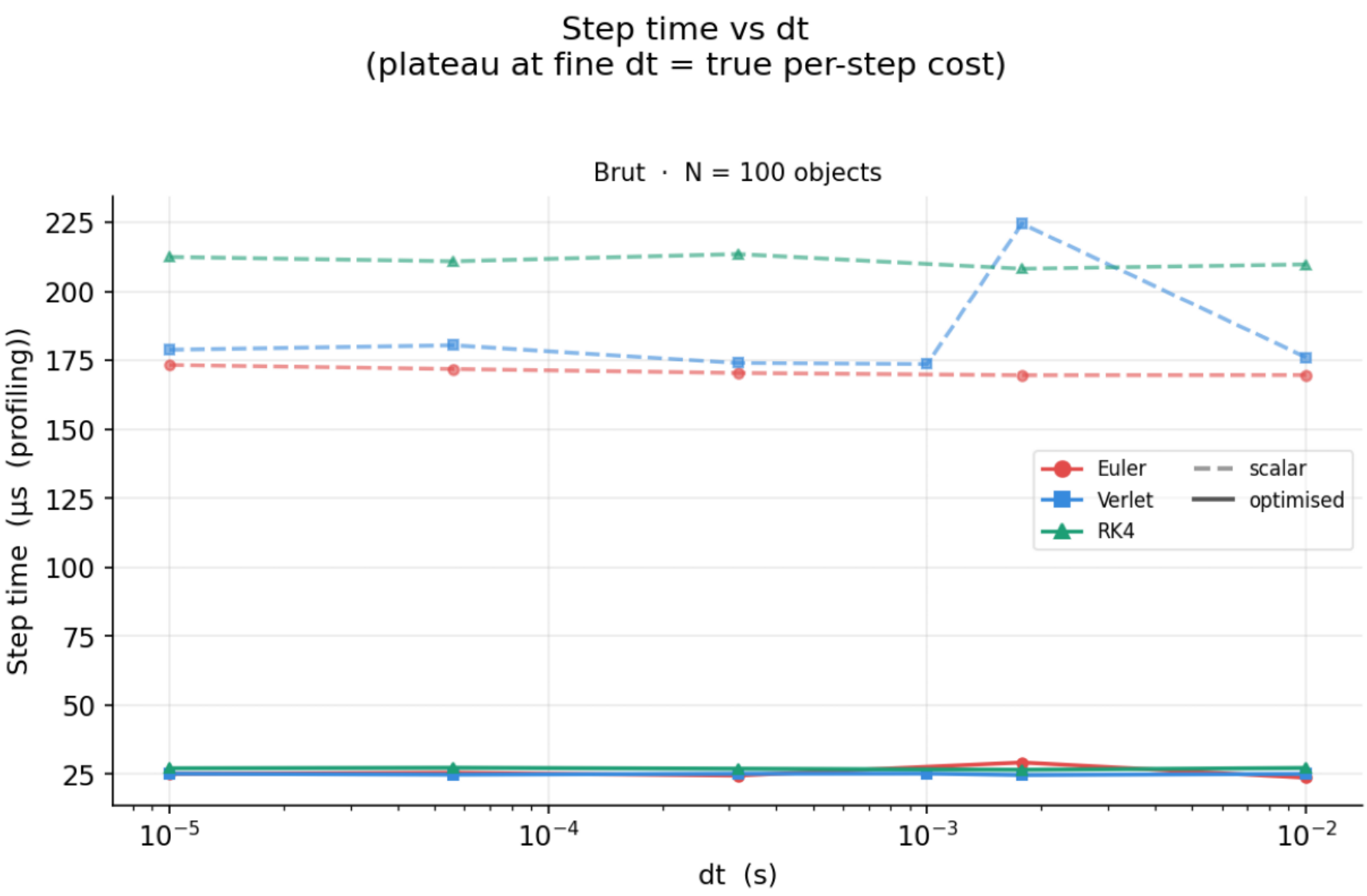
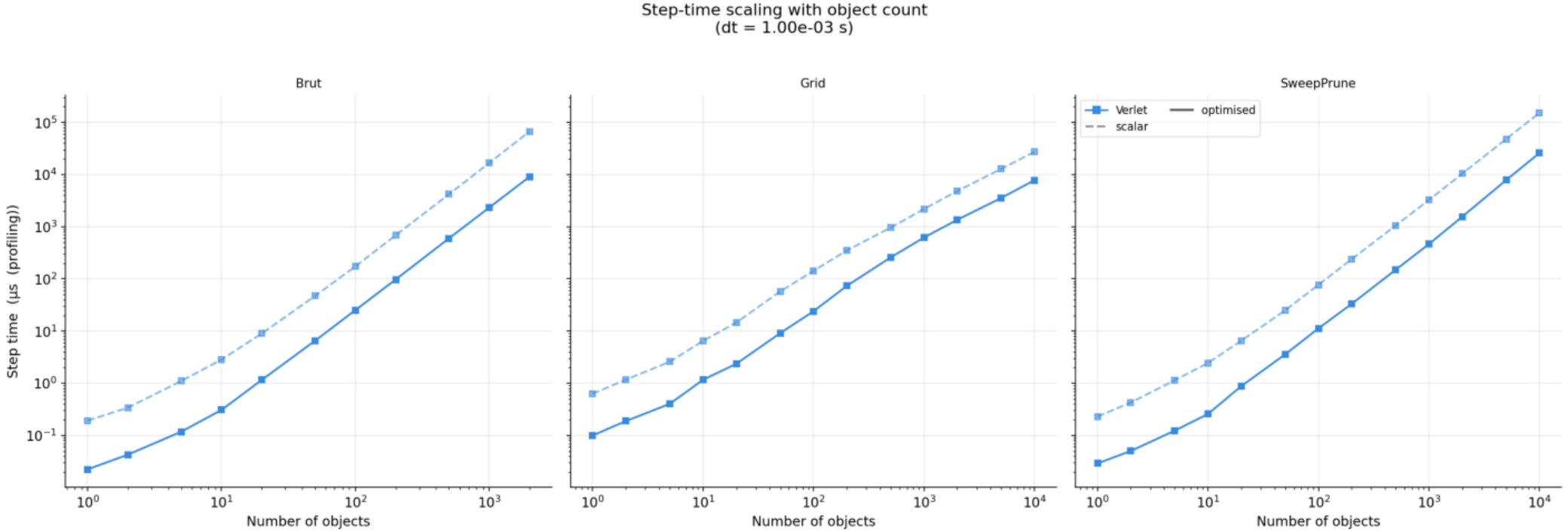
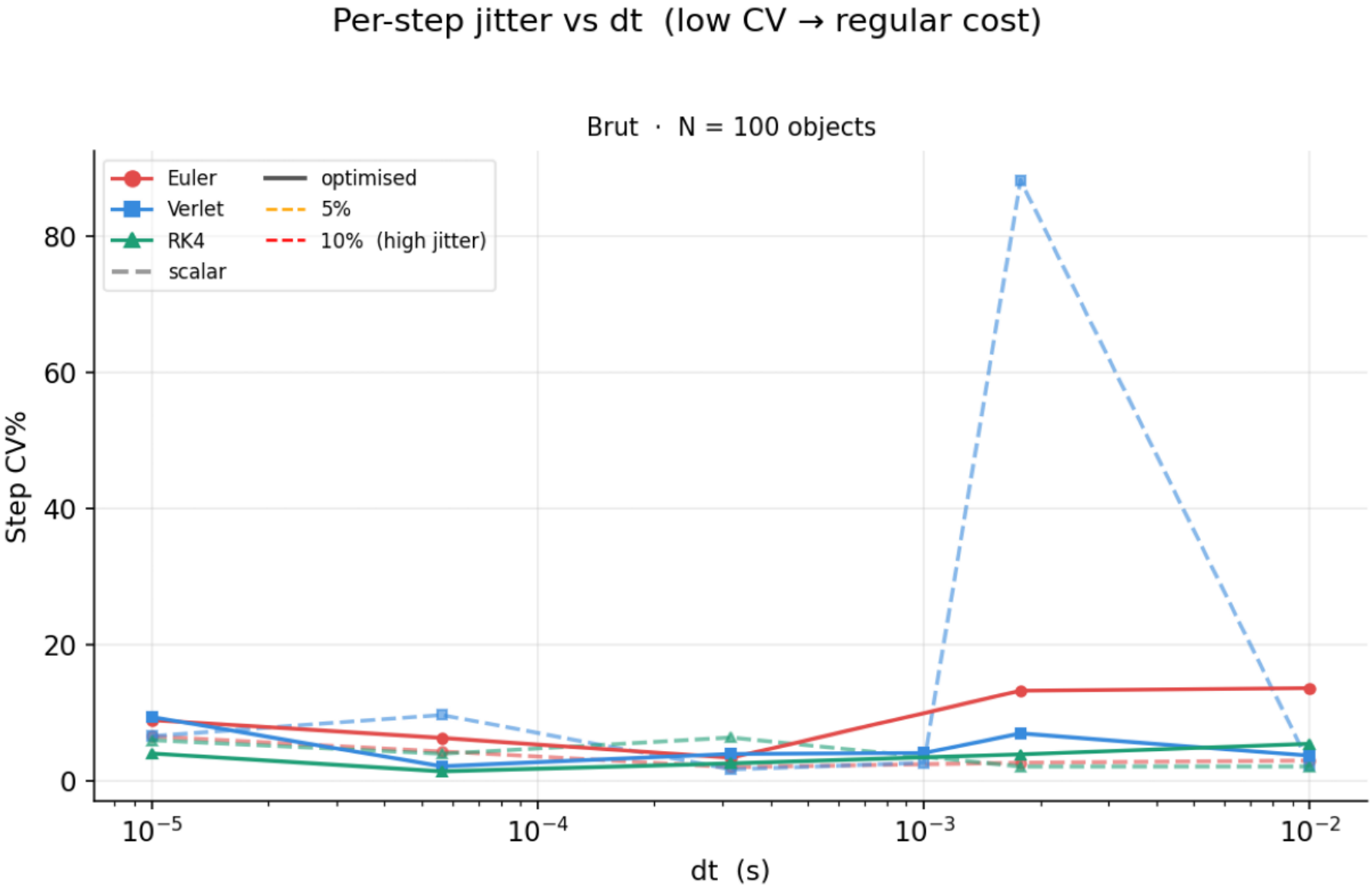
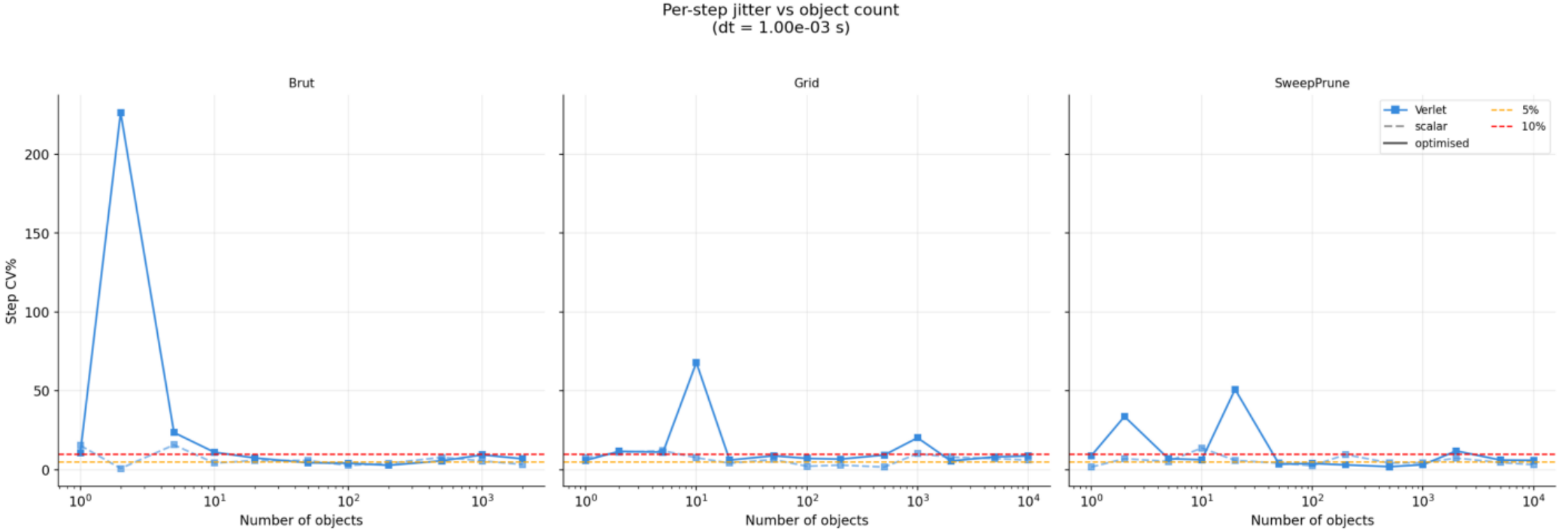

Wall time vs dt - O(1/dt) scaling expected
Brut · N = 100 objects
Euler
Verlet
RK4
scalar
optimised
Wall time (s)
dt (s)
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-5}
10^{-4}
10^{-3}
10^{-2}

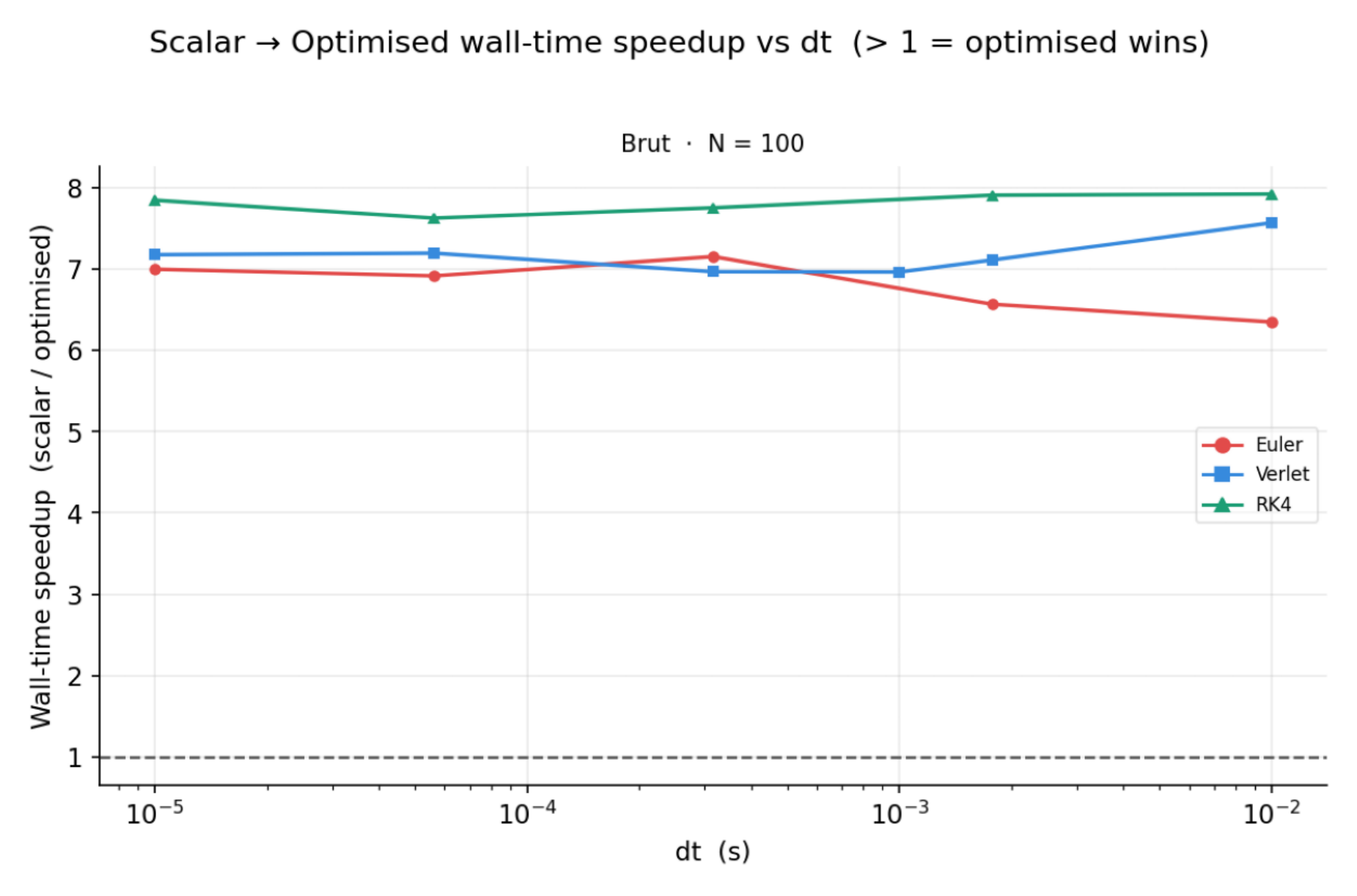

Wall-time scaling with object count
(dt = 1.00e-03 s)
Brut
Grid
SweepPrune
Verlet
scalar
optimised
Wall time (s)
Number of objects
Number of objects
Number of objects

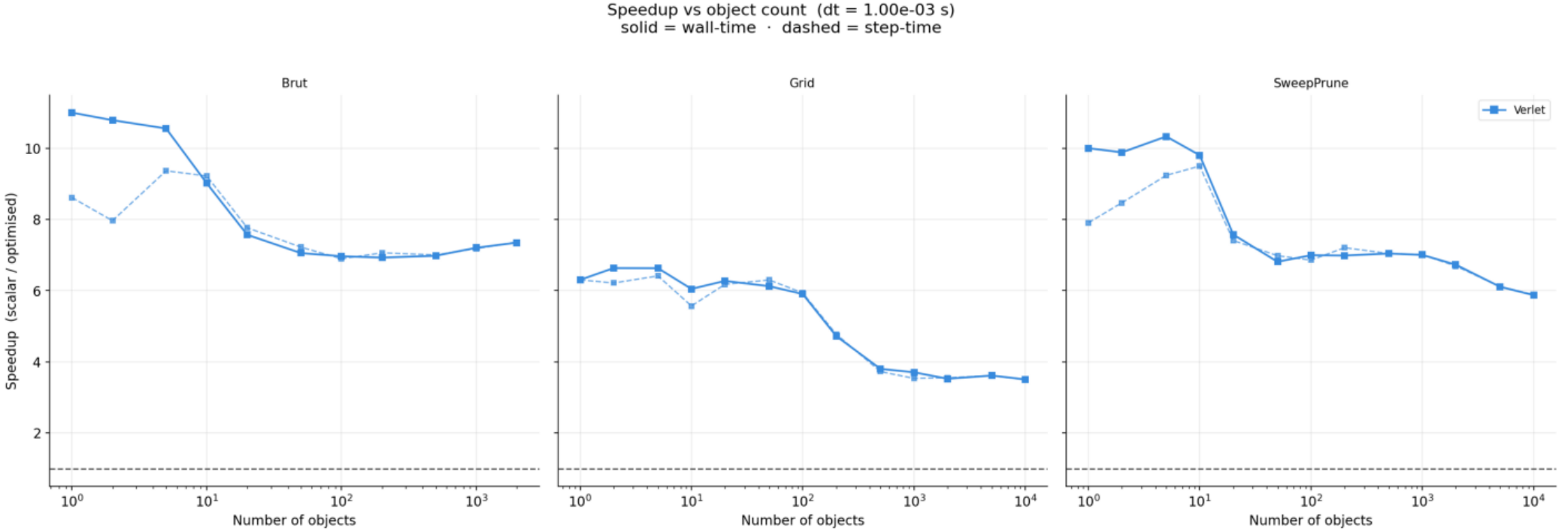

Step time vs dt
(plateau at fine dt = true per-step cost)
Brut · N = 100 objects
Step time (µs (profiling))
225
200
175
150
125
100
75
50
25
10^{-5}
10^{-4}
10^{-3}
10^{-2}
dt (s)
Euler
Verlet
RK4
scalar
optimised

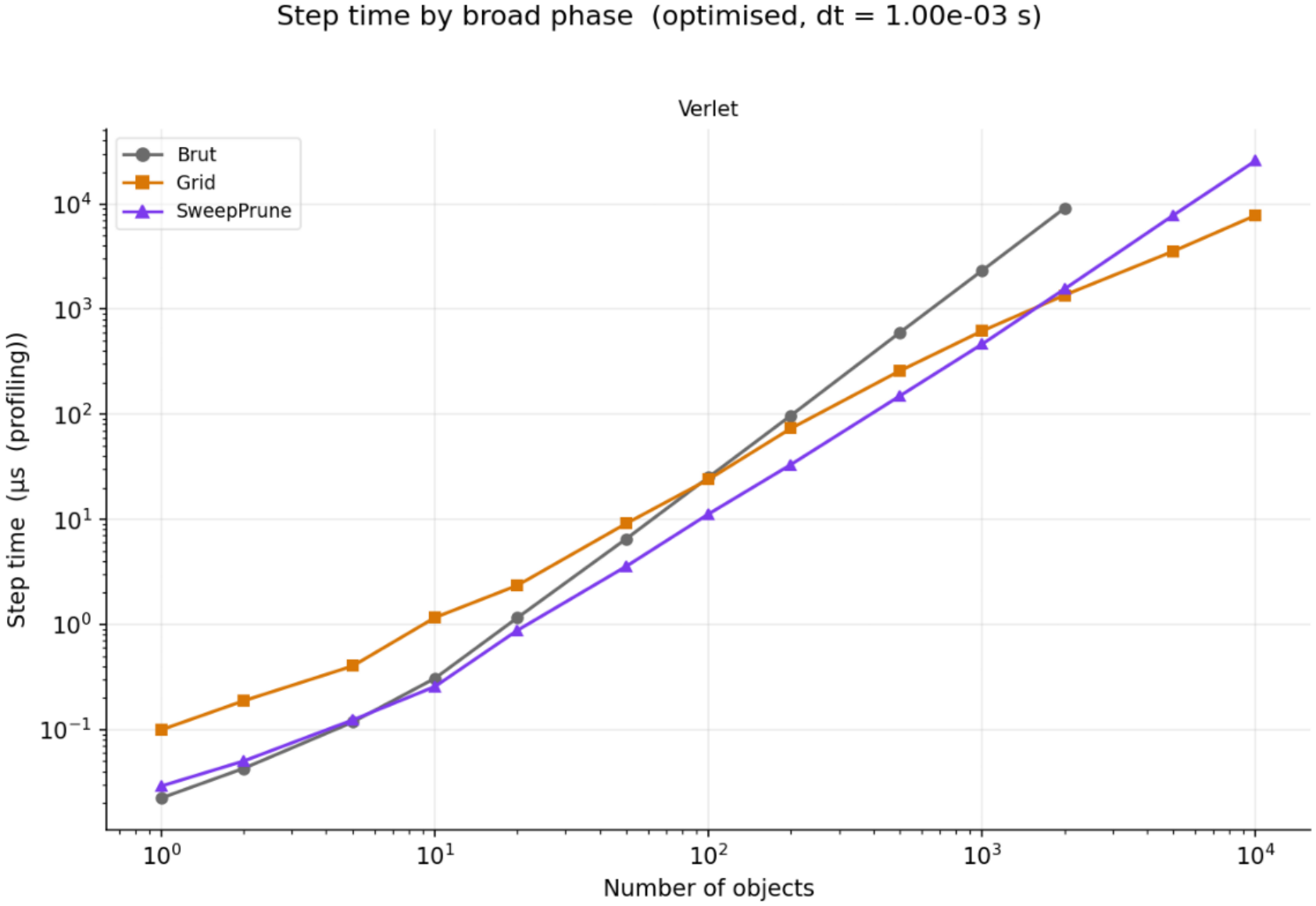

Step-time scaling with object count
(dt = 1.00e-03 s)
Brut
Grid
SweepPrune
Verlet
scalar
optimised
Step time (µs (profiling))
Number of objects
Number of objects
Number of objects

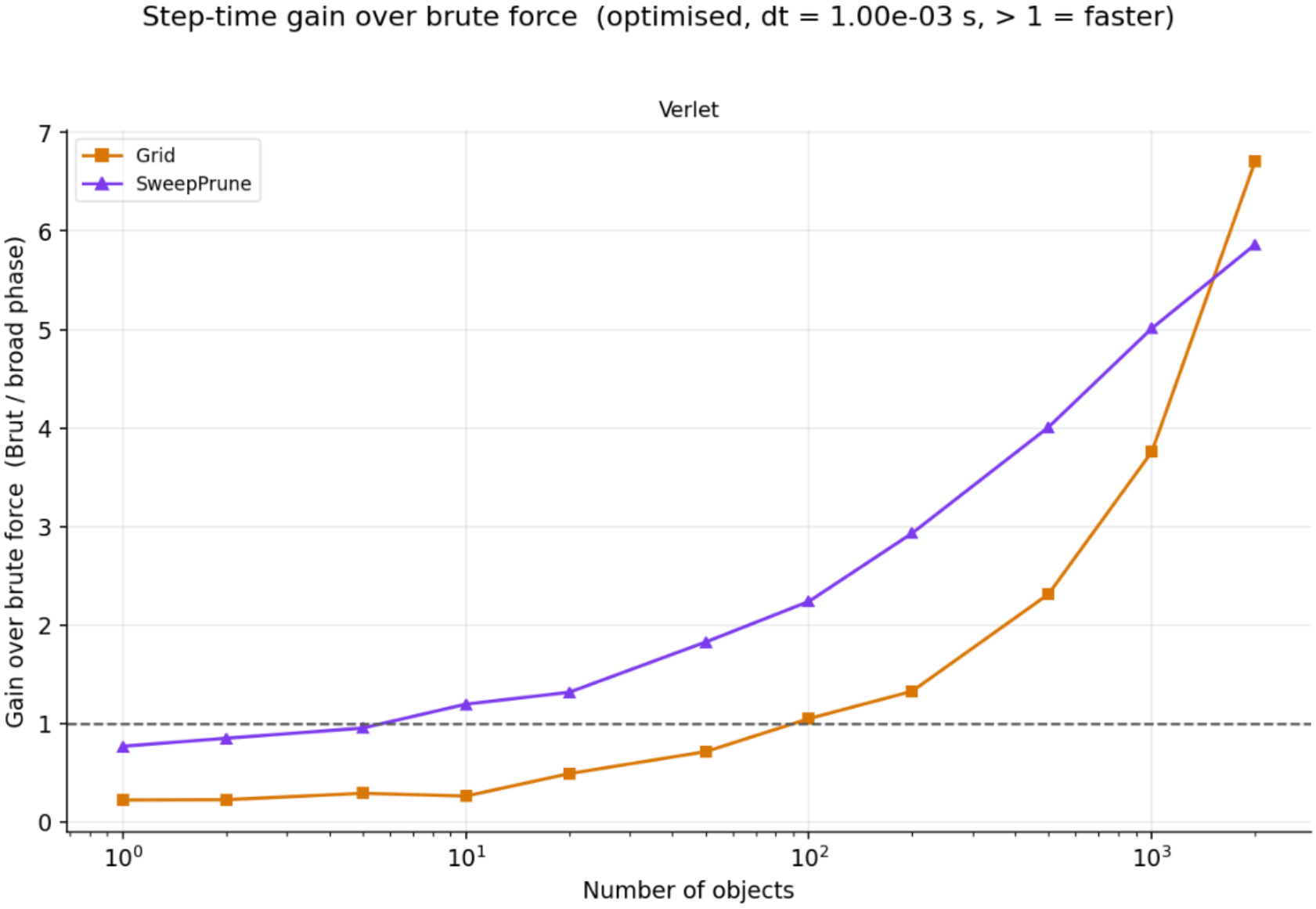

Per-step jitter vs dt (low CV → regular cost)
Brut · N = 100 objects
Euler
Verlet
RK4
scalar
optimised
5%
10% (high jitter)
Step CV%
0
20
40
60
80
10^{-5}
10^{-4}
10^{-3}
10^{-2}
dt (s)


Per-step jitter vs object count
(dt = 1.00e-03 s)
Brut
Grid
SweepPrune
Step CV%
Number of objects
Number of objects
Number of objects
Verlet
scalar
optimised
5%
10%


Scalar → Optimised wall-time speedup vs dt (> 1 = optimised wins)
Brut · N = 100
Wall-time speedup (scalar / optimised)
dt (s)
Euler
Verlet
RK4
1
2
3
4
5
6
7
8
10^{-5}
10^{-4}
10^{-3}
10^{-2}


Speedup vs object count (dt = 1.00e-03 s)
solid = wall-time · dashed = step-time
Brut
Grid
SweepPrune
Verlet
Speedup (scalar / optimised)
Number of objects
Number of objects
Number of objects


Step time by broad phase (optimised, dt = 1.00e-03 s)
Verlet
Brut
Grid
SweepPrune
Step time (µs (profiling))
Number of objects
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}


Step-time gain over brute force (optimised, dt = 1.00e-03 s, > 1 = faster)
Verlet
Grid
SweepPrune
Gain over brute force (Brut / broad phase)
Number of objects
0
1
2
3
4
5
6
7
10^0
10^1
10^2
10^3